

CHM102 Basic Organic Chemistry

Stereochemistry

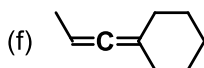
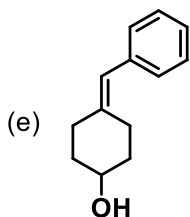
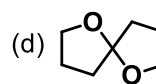
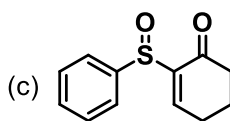
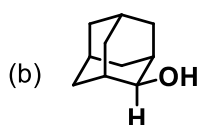
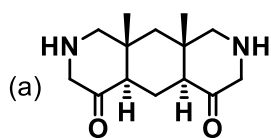
Q1. Draw all possible stereoisomers for each of the following compounds. State if no stereoisomers are possible.

a. 1-bromo-2-methylcyclohexane

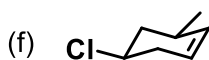
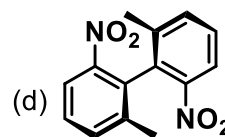
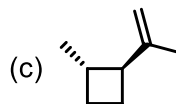
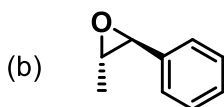
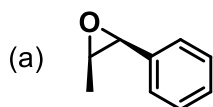
b. 1,2-dichlorocyclohexane

c. 1,2-dimethylcyclopropane

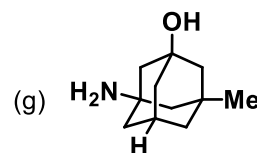
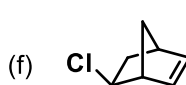
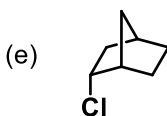
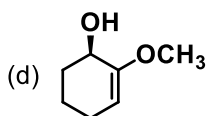
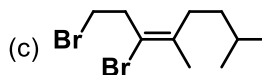
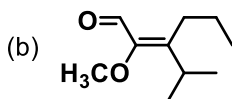
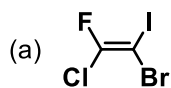
Q2. Identify whether the following molecules are chiral. Also, specify whether the compound is enantiomer or diastereomer. If not, classify the compound as achiral, meso compound.



Q3. Draw enantiomers of the following stereoisomers.



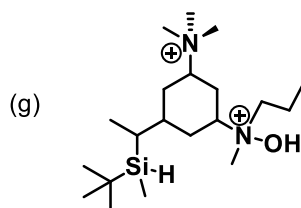
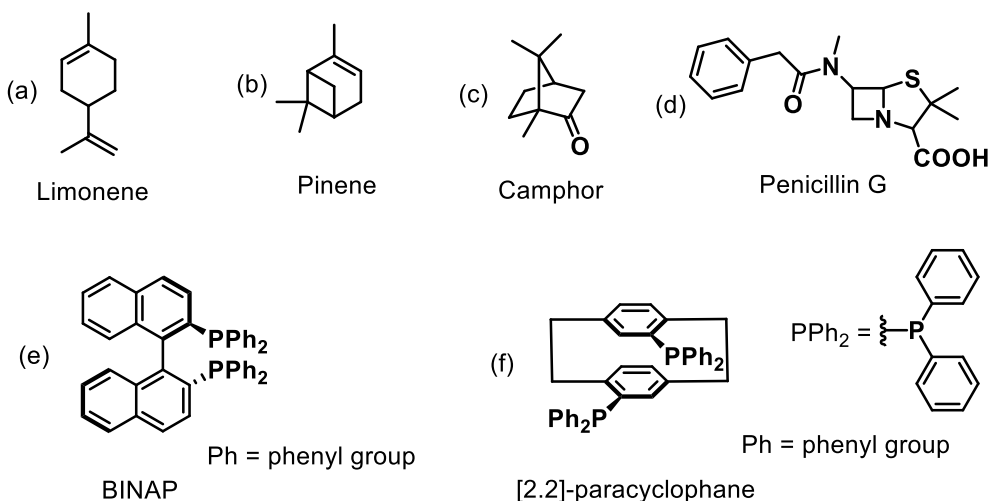
Q4. Name each of the following compounds using *R*, *S* and *E*, *Z* designations, wherever appropriate:



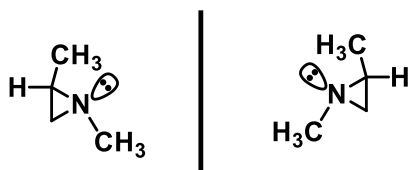
Q5. Which of the following compounds have an achiral stereoisomer?

- | | |
|-------------------------------------|------------------------------|
| (a) 2,3-dichlorobutane | (f) 2,4-dibromopentane |
| (b) 2,3-dichloropentane | (g) 2,3-dibromopentane |
| (c) 2,3-dichloro-2,3-dimethylbutane | (h) 1,4-dimethylcyclohexane |
| (d) 1,3-dichlorocyclopentane | (i) 1,2-dimethylcyclopentane |
| (e) 1,3-dibromocyclobutane | (j) 1,2-dimethylcyclobutane |

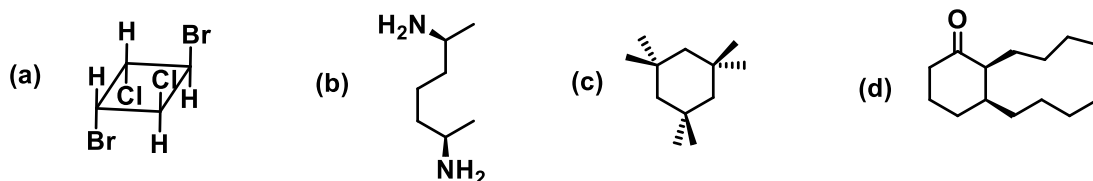
Q6. How many chiral centers are available in the following molecules? Indicate the chiral centers and/or type of chirality. Also, how many stereoisomers are possible for them?



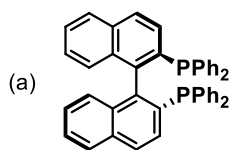
Q7. The enantiomers of 1,2-dimethylaziridine can be separated even though one of the "groups" attached to nitrogen is a lone pair. Explain.



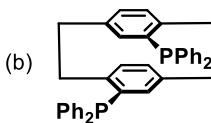
Q8. Whether the following molecule are chiral or optically active?



Q9. Identify *P* and *M* configurations (also *R* and *S* configurations) for the following compounds:

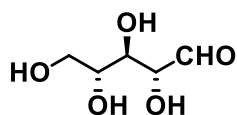


Ph = phenyl group



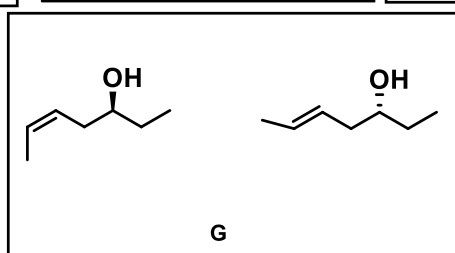
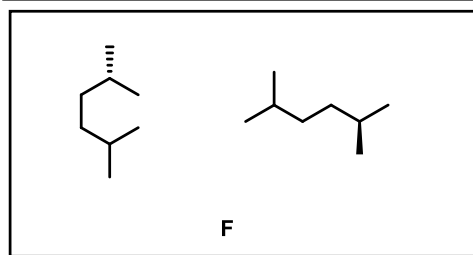
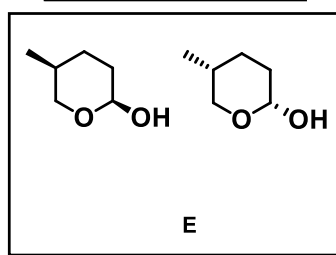
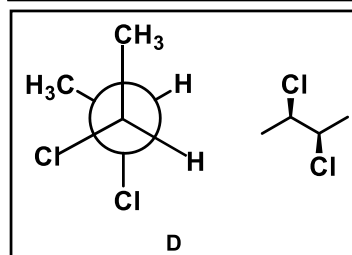
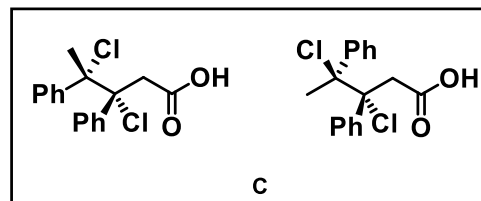
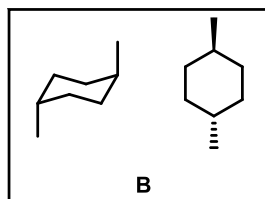
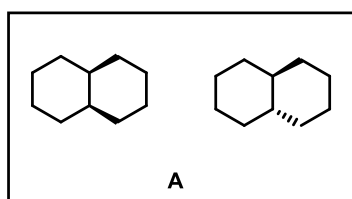
Ph = phenyl group

Q10. Draw the possible diastereomers and their enantiomeric pairs for the following molecule.

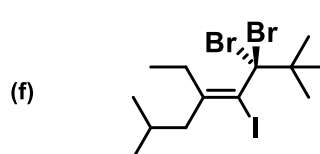
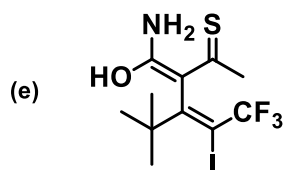
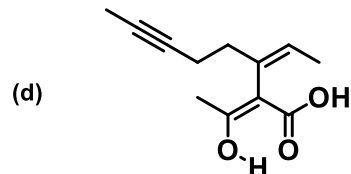
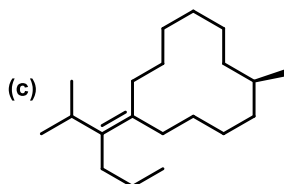
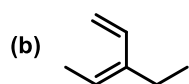
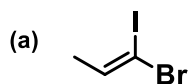


(2*R*,3*R*,4*R*)-2,3,4,5-tetrahydroxypentanal

Q11. The relationship between the following pair of molecules among the possibilities “Identical/Constitutional isomers/Enantiomers/Diastereomers” is

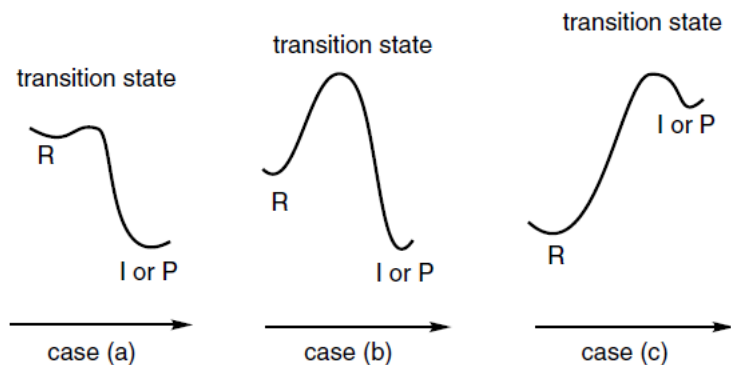


Q12. Assign absolute configurations (*E/Z*) to the following compounds. For poly-olefins, assignments should be made at each olefin. If it is not possible to assign, indicate the reason.



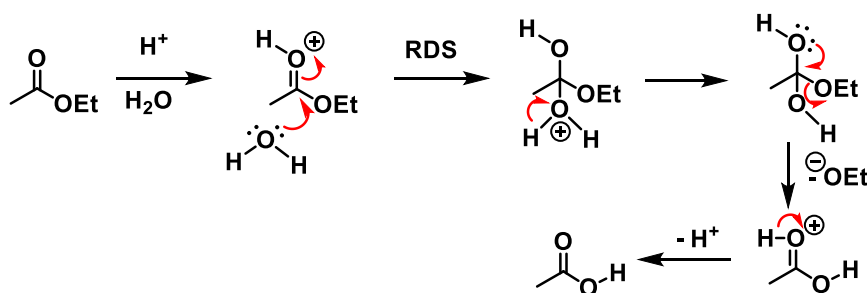
Energy profile diagrams and reactivity aspects

Q13. Account for the structure of the transition state for the following cases based on the energy profiles of the reactions (R = Reactant; I = Intermediate; P = Product):



Q14. For the following reaction, draw an energy profile diagram (based on the available data) and account for the enthalpy/entropy control in the rate determining step.

Acid-catalyzed hydrolysis of ethyl acetate in water



Q15. Construct energy profile diagram based on the known reaction mechanism for the following: (Also, indicate the rate determining step in each reaction)

- Pinacol rearrangement reaction involving $(\text{CH}_3)_2\text{CH}(\text{OH})\text{CH}(\text{OH})(\text{CH}_3)_2$ in the presence of sulphuric acid
- Dienone phenol rearrangement reaction

Q16. Indicate the dominant factor among enthalpy and entropy in controlling the following reactions:

- Base hydrolysis of ester
- E_1 , E_2 and E_1CB reactions

(c) Base mediated reaction

